SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Dave's Taxonomy: Precision (Level 3)

Question Count: 4

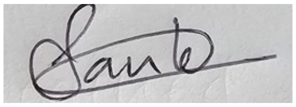
Total Marks: 40

CODE OF CONDUCT

I SANDEEP K / 192521253 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



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Cloud Computing and Big Data Analytics | CSA15 | CO3 | Total Marks: 40

1. Experiment 1: Library Book Reservation System

This SaaS experiment demonstrates a simple cloud-hosted library application for searching books and managing reservations.

- Fields: title, author, publication, year, edition, copies, rack and status.
- Implement search and availability.
- Implement reservation and taken/returned status.
- Store records in cloud database/storage.
- Test create, search, reserve, update and retrieve.

2. Experiment 2: Payroll Processing System

This SaaS application maintains employee information and calculates salary values from the required payroll fields.

- Include employee name and experience fields.
- Include basic pay, HRA, DA, TA, gross and net pay.
- Implement salary calculation logic.
- Store payroll records securely in the cloud.
- Test entry, calculation, retrieval and access control.

Cloud SaaS Architecture

3. Experiment 3: Type-2 VM – 1 CPU, 2 GB RAM, 15 GB

This experiment demonstrates virtualization by creating a guest machine on a host system with the specified resource allocation.

- Install a Type-2 hypervisor on Windows/Linux.
- Create and configure the VM image.
- Allocate 1 CPU, 2 GB RAM and 15 GB disk.
- Install and boot the guest OS.
- Verify resource allocation and operation.

Type-2 Virtualization

4. Experiment 4: VM Cloning and Previous Version

Cloning demonstrates how a configured VM can be duplicated and used as a recovery/reference environment.

- Create and configure a working VM.
- Create a clone after base configuration.
- Make controlled changes to the original.
- Load/test the cloned or previous version.
- Compare configurations and verify recovery.

Type-2 Virtualization